

Neuro-Symbolic IoT Pipeline

AI Models - Perception

Data Collection and Preprocessing

Temporal Data,
Feature Vectors

Preprocessed
tensors 

Deep learning
Recognition,
Prediction,
Anomaly Detection 

Stage 1

Symbolic Models - Reasoning

Semantic Representation and Data Federation

Learned Logical
Relationships

RDF Triples,
Ontologies 

Knowledge
Graph 

Stage 2

Real-Time Reasoning and Feedback

NLP Queries,
Explanations

Complex Events
Processing 

Feedback and
Learning 

Stage 3

Applications/
Users 

Monitoring and Adaptation:
dashboard, alerts, actions, APIs

Human Knowledge 
Experts Rules and Constraints

IoT Devices
Raw Data 

Sensor, Camera
LIDAR, UAVs, Logs
(JSON, XML)